

Changepoint Detection

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Introduction

Changepoint detection identifies the times at which the statistical properties of a time series — mean, variance, regression coefficients, or full distribution — change abruptly. Applications range from monitoring physiological signals (heart-rate or EEG state changes), to detecting regime shifts in financial and economic series, to segmenting long sensor recordings into homogeneous epochs. The output is a list of changepoint times together with the within-segment summaries between them; the methodological choices concern the change type, the detection algorithm, and the penalty controlling sensitivity.

Prerequisites

A working understanding of hypothesis testing, segmentation, and the trade-off between detecting true changepoints and false positives.

Theory

Three algorithm classes dominate:

- **Binary segmentation:** recursively detect a single changepoint, partition the series, and apply the same procedure to each segment. Fast but greedy; can miss configurations of multiple nearby changepoints.
- **PELT** (Pruned Exact Linear Time): finds the optimal segmentation under a penalty in $O(n)$ when the penalty satisfies a regularity condition. The standard modern method.
- **Segment Neighbourhood:** dynamic programming for the optimal partition under a fixed maximum number of changepoints.

The penalty (BIC, MBIC, Hannan-Quinn, or AIC) controls how many changepoints are reported; a larger penalty yields fewer changepoints. Different change types have different test statistics: mean changes use the likelihood-ratio test under a Normal model; variance changes use the same with σ as the parameter; meanvar combines both; regression changepoints generalise to coefficient breaks.

Assumptions

Independent observations within segments (or correctly specified within-segment dependence structure); the change type matches the underlying signal (mean shifts vs variance shifts vs regression breaks).

R Implementation

```
library(changepoint)

set.seed(2026)
x <- c(rnorm(100, 0, 1), rnorm(100, 3, 1), rnorm(100, -1, 1))

# Mean changes via PELT
cp <- cpt.mean(x, method = "PELT", penalty = "BIC")
cpts(cp)
plot(cp)
```

```
# Variance changes
cp_v <- cpt.var(x, method = "PELT")
cpts(cp_v)
```

Output & Results

`cpt.mean()` returns a `cpt` object with the changepoint times accessible via `cpts()`, the within-segment means via `param.est()`, and a plot showing the detected segments. The simulated three-segment structure is recovered with changepoints at times 100 and 200.

Interpretation

A reporting sentence: “PELT with BIC penalty detected two changepoints at times 100 and 200, recovering the simulated three-segment mean structure (segment means 0.04, 2.95, -0.97); a sensitivity analysis with MBIC and Hannan-Quinn penalties yielded the same changepoints.” Always report the algorithm, the penalty, and a sensitivity check across penalties.

Practical Tips

- Choose the right change type: `cpt.mean` for mean shifts, `cpt.var` for variance shifts, `cpt.meanvar` for both, regression-based methods for coefficient breaks.
- The penalty controls false positives; try several (BIC, MBIC, Hannan-Quinn) and check that conclusions are robust.
- For non-stationary noise within segments, model the noise (ARIMA) first and apply changepoint detection to the residuals.
- Bayesian alternatives (`bcp::bcp`) give per-time changepoint posterior probabilities, which are more informative than a hard list of detected changepoints.
- PELT is offline — it requires the full series. For online (real-time) detection, use CUSUM, Page-Hinkley, or Bayesian online methods.
- For multivariate changepoint detection, `ecp` and `InspectChangepoint` extend PELT-style methods to vector time series.

R Packages Used

`changepoint` for `cpt.mean()`, `cpt.var()`, `cpt.meanvar()`, and PELT/Binary Segmentation algorithms; `bcp` for Bayesian changepoint detection with posterior probabilities; `ecp` for non-parametric multivariate changepoints; `cpm` for online changepoint methods.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Time series basics